

Lung cancer

Lung cancer is a type of cancer that starts in the lungs.

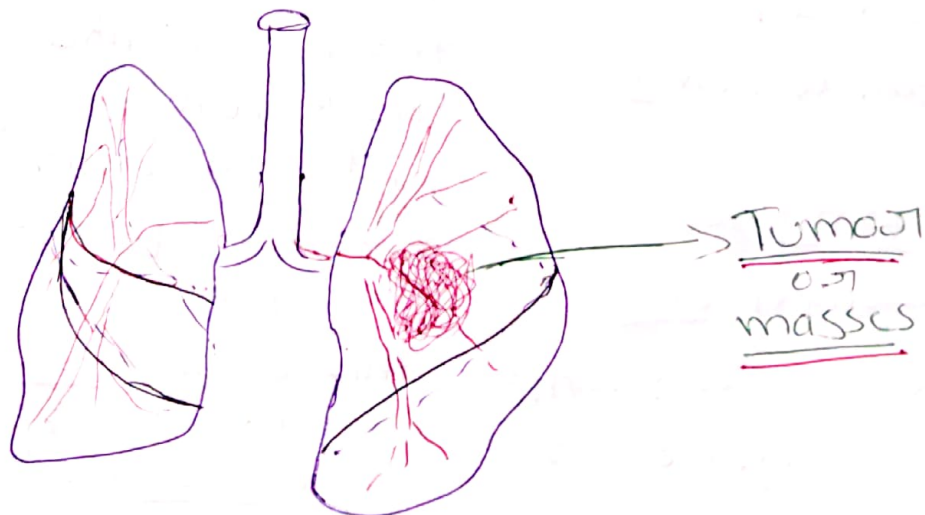
when cells in the Lungs mutate & grow uncontrollably,

forming tumors that destroy healthy lung tissue and can spread throughout the body.

It caused by uncontrolled cell division in your lungs

Your cells divide and make more copies of themselves as a part of their normal function.

Damaged cells dividing uncontrollably create masses or tumors



Lung Cancer

85%

Non-small cell lung cancer

15%

Small cell lung cancer

15%

Adenocarcinoma

mucin → is responsible for keeping the lungs moist



In this mature cell has the function of secreting mucin.

Adeno meaning coming from a gland

This occurs 40% of the time in all non-small cell lung cancers

This is going to affect women because it ~~actually~~ affects both smokers & nonsmokers

Squamous cell carcinoma

Desmosome



This cells creates a barrier btw airways & rest of the body

keratin is also in the cells
this is a lot like the keratin you find in our hair & serves a protective function

this occurs 30% of the time & in males this time

Males who have a history of smoking

High smoking

large cell carcinoma



It occurs 10% in male smokers

carcinoid



Flash shaped cell

Nerve

Some Nerve when it receives signals from these nerves

It releases hormones

This happens 5% of time

Interesting Male & female have an equal risk of developing carcinoid cancers
no direct link to

Risk factors

- ☐ Smoking
- ☐ Secondhand smoke
- ☐ exposure to radon
- ☐ Family history lung cancer
- ☐ HIV infection
- ☐ Harmful chemicals
- ☐ Air pollution

Symptoms

- ☐ Chronic cough
- ☐ chest pain
- ☐ trouble breathing
- ☐ Coughing up blood
- ☐ Hoarseness
- ☐ loss of appetite
- ☐ Trouble swallowing
- ☐ weight loss
- ☐ tiredness
- ☐ Facial or neck swelling

If have any lung cancer doctor will need to determine the stage or progression of the disease to help plan your treatment

Staging

Non-small cell lung cancer

Stages

Occult: in this stage cancer cells are found only in lung fluids

S0: abnormal or cancer cells found in the lining of the airways

S1: A small tumor is found

S2: the cancer has grown or spread to nearby lymph nodes or non-lung tissues

S3(A): The tumor has grown to any size in addition cancer is found in lymph nodes on the same side of the chest
(as)
in any nearby organs

S3(B): It is similar to 3(A) but in 3(B) cancer is found in lymph nodes on the other side of the chest or lung

S4: The Tumors might be found in both lungs - additionally cancer may have spread to distant organs

Small cell lung cancer

Stages

Limited: in the limited stage cancer is found in one side of the chest or lungs only

extensive: The cancer has spread to the other side of the chest or lungs or to other or distant organs

Pack - Year:

Pack year is the number of cigarette packs smoked each day multiplied by the number of years a person has smoked for

so ~~one~~ more pack years of smoking history in their life

Pack years is a way to assess how much a person is smokes in their life to

To calculate pack years multiply the number of cigarette packs a person smoked per day by the number of years of cigarette use for
for ex

a person who smokes one pack per day for 20 years has a 20 pack year history.

if someone smokes half a pack for 40 years that would also be 20 pack years history

$$\begin{array}{c} \boxed{1 \text{ Pack}} \\ \text{Per day} \end{array} \times \begin{array}{c} \boxed{20 \text{ Years}} \\ \text{years} \end{array} = 20 \text{ Pack Years}$$

$$\begin{array}{c} \boxed{5 \text{ Packs}} \\ \text{Per day} \end{array} \begin{array}{c} \text{half pack} \\ \boxed{1/2} \end{array} \times \begin{array}{c} 40 \\ \text{years} \end{array} = 20 \text{ Pack Years}$$

Diagnosis and screening of lung cancer

doctors use several methods and tests for diagnosing lung cancer like including

• Patient history • family • work • Medical

• Smoking History

~~Physical~~
exam

Screening and Diagnosis

Physical exam

Chest X-ray

CT, PET or MRI

Transthoracic needle biopsy

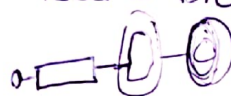
Bronchoscopy

Running in

computer tomography,

Positron emission tomography,

Magnetic resonance imaging scans



Passes a thin flexible tube known as a bronchoscope from mouth or nose through the windpipe into breathing passages of the lungs. The scope has a tiny video camera & light on its tip.

In recent group recommend low dose chest CT screening
People who are 55 to 74 year old

Small sample your lung tissue where cancer is suspected is known as lung biopsy